Problem Solving in Geometry (V)
17.02.2024

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1. Let ABC be a triangle and let D be the orthogonal projection of A on the line BC. Let K be a point on the segment AD such that $AD = 3KD$. Let O be the circumcenter of triangle ABC and let M and N be the midpoints of sides AC and AB respectively. The lines KO and MN meet at a point Z and the perpendicular at Z to OK meets lines AB, AC at X and Y respectively. Show that $\angle XKY = \angle CKB$. (RMM2018SLG1)

2. Let BM be a median in an acute-angled triangle ABC. A point K is chosen on the line through C tangent to the circumcircle of $\triangle\ BMC$ so that $\angle KBC = 90^\circ$. The segments AK and BM meet at J. Prove that the circumcenter of $\triangle BJK$ lies on the line AC. (RMM2019SLG1)